Youssef Essam

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Summary

Software Engineer with a Mechatronics Engineering degree, specializing in full stack development (**React**, **TypeScript**, **Node.js**) and robotics/imaging systems. Experienced in building high-performance control software, real time data pipelines, and embedded applications using **Python**, **C++**, **Qt/QML**, and **ROS2**.

Education

Mechatronics Engineering — *Feb 2025*

Bachelor's Degree - Yildiz Technical University - GPA: 3.02 / 4

Experience

Software Engineer — Istanbul, Turkey, In Office
Technomind Digital Systems — *June 2025 - Current*

• Built and maintained the web control interface for a digital pathology device using **React**, **TypeScript** and **Tailwind CSS**, and integrated **RESTful APIs** for device communication.

• Implemented backend microservices using **Python (Flask)** and **TypeScript (Express.js)** to support real time device control and data management.

• Increased scanning throughput by **818%** by identifying pipeline bottlenecks and redesigning the workflow with **asynchronous execution**, **queues**, and **multiprocessing** to parallelize processing stages and reduce latency.

• Refactored legacy codebases into clean, modular components.

Software Developer — Istanbul, Turkey, Hybrid
Letna Marine Underwater Engineering Solutions — *Feb 2023 - June 2025*

• Developed and maintained features for Pyrot, a production level fork of **QGroundControl** built with **Qt (C++/QML)**, improving usability for **20+ ROV operators**.

• Implemented new ROV control modes in **C++**, improving navigation stability and reducing operator error.

• Developed a Python based **computer vision** system for real time distance estimation using dual laser alignment and image processing, achieving **95.5% measurement accuracy**.

• Developed custom **QML** HUD overlays to improve **UI/UX** by displaying critical sensor data without obstructing the operator's field of view.

Software Developer — Istanbul, Turkey, Hybrid
Lagari UAV team — *Jan 2022 - June 2022*

• Developed autonomous flight logic and computer vision systems for the 2022 **TEKNOFEST** International UAV Competition as part of the software team.

• Developed a real time target detection and geolocation pipeline using **Python** and **OpenCV** to identify ground targets from UAV imagery.

• Integrated autonomous navigation logic into the UAV control stack to enable precision payload delivery to detected targets.

Projects

Portable Knee Rehabilitation Robot - TUBITAK 2209-B Support Programme — *Github: Raspberry Pi Code | Qt Android App Code*

Technologies: **Python, ROS2, Qt (C++/QML)**
• Designed a **ROS2** based control architecture on **Raspberry Pi** for real time sensor integration and motor control.
• Developed custom PID controllers to enable adaptive knee rehabilitation exercises.
• Implemented three rehabilitation modes (Training, Exercise, Impedance), controllable via Bluetooth commands.
• Developed an Android compatible **Qt/QML GUI** for real time feedback and knee angle visualization.
• Implemented **Bluetooth RFCOMM communication** for reliable command and sensor data transfer.

Personal Portfolio Website — *https://youssefessam.com*

Tech Stack: **React.js**, **Vercel**
• Built and deployed a responsive portfolio website using **React.js, Tailwind CSS** and **Vercel** to showcase full stack, embedded, and robotics projects with optimized performance and clean UI architecture.

Skills

Programming languages : **Python**, **C++**, **JavaScript**, **TypeScript**, **HTML**, **CSS**
Frameworks & Libraries: **React.js**, **Next.js**, **Tailwind CSS**, **Node.js**, **Express JS, Flask, Qt (C++/QML)**
Tools & Technologies: **Git**, **GitHub**, **Linux**, **ROS2**, **Arduino**, **Raspberry Pi**, **Figma**
Languages: **Turkish (C1)**, **English (C1)**, **Arabic (C1)**